

Rationalising the denominator

Rationalise the denominator,

$$1) \frac{4 + \sqrt{3}}{5 + \sqrt{6}} = \frac{20 - 4\sqrt{6} + 5\sqrt{3} - 3\sqrt{2}}{19}$$

$$2) \frac{4 + \sqrt{3}}{4 + \sqrt{7}} = \frac{16 - 4\sqrt{7} + 4\sqrt{3} - \sqrt{21}}{9}$$

$$3) \frac{2 + \sqrt{6}}{5 + \sqrt{3}} = \frac{10 - 2\sqrt{3} + 5\sqrt{6} - 3\sqrt{2}}{22}$$

$$4) \frac{5 + \sqrt{2}}{4 + \sqrt{2}} = \frac{18 - \sqrt{2}}{14}$$

$$5) \frac{5 + \sqrt{2}}{4 + \sqrt{6}} = \frac{20 - 5\sqrt{6} + 4\sqrt{2} - 2\sqrt{3}}{10}$$

$$6) \frac{3 + \sqrt{5}}{4 + \sqrt{5}} = \frac{7 + \sqrt{5}}{11}$$

$$7) \frac{2 + \sqrt{7}}{5 + \sqrt{2}} = \frac{10 - 2\sqrt{2} + 5\sqrt{7} - \sqrt{14}}{23}$$

$$8) \frac{5 + \sqrt{5}}{5 + \sqrt{2}} = \frac{25 - 5\sqrt{2} + 5\sqrt{5} - \sqrt{10}}{23}$$

$$9) \frac{4 + \sqrt{7}}{3 + \sqrt{3}} = \frac{12 - 4\sqrt{3} + 3\sqrt{7} - \sqrt{21}}{6}$$

$$10) \frac{4 + \sqrt{6}}{3 + \sqrt{6}} = \frac{6 - \sqrt{6}}{3}$$